		$x = 0.109$ m
10	**C**	At maximum speed, engine force = drag force of kv. Power of boat, P = (engine force) v = kv^2 Hence, $\dfrac{P_{\text{one engine}}}{P_{\text{two engines}}} = \left(\dfrac{v_{\text{one engine}}}{v_{\text{two engines}}}\right)^2$ $\dfrac{32}{64} = \left(\dfrac{v_{\text{one engine}}}{14}\right)^2$ $v_{\text{one engine}} = 9.9 \text{ m s}^{-1}$

11	A	There are only two forces acting on the person, force of cage on him, *R* and his weight, *W*.